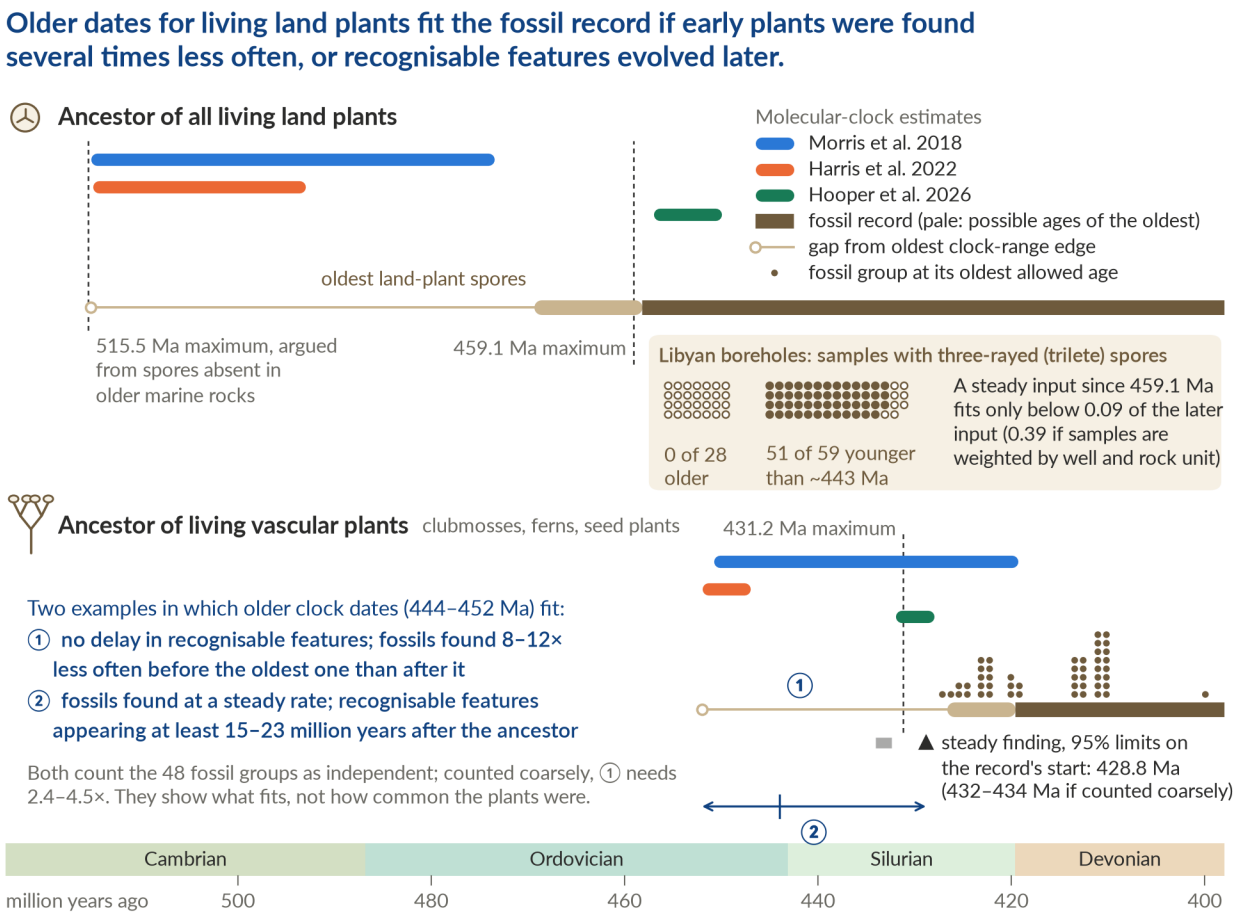
When did living land plants arise? What their fossils require

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Graphical abstract. Coloured bars: molecular-clock estimates of each ancestor's age. Brown band: the fossil record, its pale start spanning the possible ages of the oldest finds; thin line: the gap back to the oldest edge of the clock ranges. Vascular-plant dots sit at each group's oldest allowed age, stacked for display; the Libyan grids are sample counts, not dates. Examples 1 and 2 were chosen after seeing the data and show compatibility, not 95% limits; the 95% limits assume a steady recording rate. All describe the fossil record; the ancestor itself can be older.

The puzzle

Every forest, grassland and crop descends from a few small plants that colonised the land hundreds of millions of years ago. Those plants changed the air, the soils and the climate. Yet scientists disagree by tens of millions of years about when the ancestor of today's land plants lived.

A *molecular clock* compares the DNA of living plants and works out how long their lineages have been diverging. It dates the shared ancestor of living species, which may be younger than the first plants on land. Clocks need fossils to set their time scale, above all *maximum ages*: arguments that a much older group would probably have left recognisable fossils in the rocks examined. Move the maximum and the clock's answer moves with it. Three recent clock studies place the ancestor of all living land plants 515–474 million years ago (Ma) or 457–450 Ma, depending on their maximums, and the ancestor of living *vascular* plants, which have internal plumbing, between about 452 and 428 Ma (Morris et al. 2018; Harris et al. 2022; Hooper et al. 2026).

Why early plants are hard to date

The oldest traces of land plants are tough, microscopic spores. Spores appear in the Middle Ordovician, fragments of spore capsules in Late Ordovician Oman, and branched plants, often only millimetres tall, in

the Silurian. Missing fossils in older rocks can mean the plants were absent, or that small, woodless and perhaps rare plants were seldom preserved, found and recognised. A maximum age has to argue for the first reading.

What we did

We traced the three maximum ages that divide the clocks (515.5 and 459.1 Ma for land plants, 431.2 Ma for vascular plants) to their original fossils, ages and arguments. We modelled spore records from Libyan boreholes, where none of 28 samples older than about 443 Ma reports three-rayed (*trilete*) spores while 51 of 59 younger samples do; 10 of the older samples are broadly dated and may reach back beyond 459 Ma. From how 48 dated groups of vascular-plant fossils are spread through time, we calculated how far back that record could reach under several histories of how often plants left fossils that were later found.

What we found

1. **The oldest maximum (515.5 Ma) rests on a qualitative argument.** Older marine rocks preserve tough algal cysts and no land-plant spores, so its authors read the spores' absence as real. How many spores reached those seas, and how hard anyone looked, remain unmeasured.
2. **The Libyan rocks limit sustained, well-delivered spore input.** A steady flow of spores into these rocks from 459.1 Ma fits only below about a tenth of the later flow (0.091); lower input, poor delivery or plants living elsewhere all fit. This treats samples from one well as sharing a well effect and otherwise independent; weighting samples by well and rock unit instead gives an exploratory figure near 0.4. The result covers mainly the later Ordovician in one polar basin.
3. **Vascular plants can be older if early fossils were scarcer, or recognisable later.** At a steady recording rate, the 95% older limit on their body record is about 429 Ma, or 432–434 Ma if the finds are counted as fewer independent places. A retrospective scenario that halves the earlier rate reaches only 431.0 Ma. Dates of 444–452 Ma fit if such fossils were found 8 to 12 times less often before the oldest find than after it (2.4 to 4.5 times if counted cautiously), or at a steady rate if recognisable bodies appeared at least 15–23 million years after the ancestor.
4. **One find can move the limit by 20 million years.** Counting the tiny Late Ordovician spore-capsule fragments from Oman moves the limit for land-plant bodies back by about 20 million years.

What it means

The vascular-plant record leaves room for older ancestors if early plants were several times scarcer in the record, or gained their recognisable features later. The Libyan spores leave room for older ancestors whose spores reached that basin sparsely or not at all. Plants that were uncommon, local, small or seldom searched for would leave exactly such a record; our calculations give the size of the requirement, and its biological plausibility remains open. Clock builders can state the fossil, geographical and recording assumptions behind each maximum and test how alternative maxima change their dates. In short: older dates for living land plants fit the fossil record if early plants were found several times less often, or recognisable features evolved later.

How this work was done

Two AI systems, Claude (Anthropic) and Codex (OpenAI), did the research with Barry Brook, who set its direction; independent reviews by GPT-6 Pro (OpenAI) shaped it. Every number traces to a file and script in the public repository.

Sources

Harris, Brogan J., James W. Clark, Dominik Schrempf, Gergely J. Szöllősi, Philip C. J. Donoghue, Alistair M. Hetherington, and Tom A. Williams. 2022. "Divergent Evolutionary Trajectories of Bryophytes and Tracheophytes from a Complex Common Ancestor of Land Plants." *Nature Ecology & Evolution* 6 (11): 1634–43. https://doi.org/10.1038/s41559-022-01885-x.

Hooper, Jack, Annabel M. Worth, Charles H. Wellman, James W. Clark, and Philip C. J. Donoghue. 2026. "Reconstructing Spore Evolution Recalibrates the Timescale of Land Plant Diversification." *Nature Plants*. https://doi.org/10.1038/s41477-026-02432-3.

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